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A Neural Network Framework for Estimation of Three Dimensional Object Location in Small Rooms Using Acoustic Echoes

Thesis proposal submitted in partial fulfilment to opt for the degree of

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Author

Esteban Roberto Padilla Padilla

Supervisor

Ing. Kervin Sanchez Herrera, M.Sc.

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Abstract

This document presents the thesis topic proposal for obtaining the Master's degree in Computer Science: Emphasis in Computer Science at the Tecnológico de Costa Rica. The proposed topic is centered in the idea of the potential use of machine learning techniques to analyze the patterns in acoustic echoes produced inside small rooms to estimate the location of certain objects in a tridimensional space. Two neural networks architectures are proposed for this task, a Convolutional Neural Networks that despite being an architecture mainly for images is commonly used in literature for acoustic inference, and an Echo State Machine as a new architecture with potential to address directly the acoustic signals. The effect of beamforming is also considered as one of the elements to investigate as a solution to the multi-path acoustic superposition. The design for the research is presented, the required experimental setup is discussed as well as the methodology to measure, process and evaluate the results, all this with the objective to evaluate the potential use of sound echoes in the audible frequency as a mean to locate object in an enclosed room.

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Introduction

Robots are expected to navigate in multiples and different environments, in order to achieve this we need to allow them to have situational awareness and autonomy to make proper decisions. Computer vision aims to enable machines to get an interpretation of the world that surround them. Processing images and videos robots can extract information to recognize objects geometries, textures, materials, etc. Although it is true that with visual analysis, the machine can create internal models that contain a vast knowledge of the environment, this doesn't mean that its perception is complete. Robotic perception give the ability to a machine to understand its surroundings to the degree that could safely interact with the world [12], for such understanding other techniques are required as well as computer vision to get a complete sensory experience of the environment. Acoustic sensing and analysis can enhance the perception of machinery following an approach similar to that seen in nature with animals that use binaural acoustic perception, the environment can be analyzed to detect the source of a particular sound and focus the visual attention to that particular region, but, this capability is not only an extension,

in some conditions the animal can entirely relies on the acoustic information to operate, this is the case of echolocation employed by bats, birds, dolphins and wales, they ear the echoes of acoustic chirps to gain spatial perception of the surrounding environment and this enable them to navigate in complex layouts [38].

Traditional computer vision techniques that employ sensing devices such as cameras and lidars provide high-fidelity information to navigation systems, but they become unreliable in scenarios with poor light or presence dust, smoke, and fog [32]. Another aspect to consider is the cost of the sensing device, here the acoustic perception offers an advantage due the low cost of commercial electret or MEMS microphones which are inexpensive compared to optical cameras and lidars; then become desirable its use as an alternative to traditional sensors. Recent developments in machine learning as the ones shows in [2; 25; 32] demonstrated the big potential to use neural network to infer spatial and geometrical information of objects in the surrounds by just analyzing acoustic echoes in the audible spectrum. The manufacturing sector under the industry 4.0 transformation is increasing in the adoption of automation and smart monitoring technologies and so it has become an important field of application of these new methods, where the supervision of machinery during operation, especially the machines allocated in production floors using robotic arms and computer numerical control actuators intended to manipulate objects in confined spaces [23]. In such conditions the acoustic sensing is a good approach to get an insight of the overall status of the system due to the nature of the Room Impulse Response as a transfer function that comprehends information of the environment setup. The acoustic response of a room can be use together with artificial neural networks like convolucional neural networks or recurrent neural networks to learn from this response and be capable of recognize characteristics and properties of the room echoes to recognize spaces and objects in it. Thus, acoustic perception together with neural networks can improve the perception of machinery

when it operates in complex spaces.

1.1. Problem

The use of acoustic signal to approximate the size and shape of objects was considerably improved by the adoption of machine learning techniques. An important part of the work in this field was develop with the scope of indoor navigation systems where the objective is to enhance the perception of robots using acoustic analysis to build low resolution deep images of an unknown environment. A different point of view is formulated and presented in this work, one in which we know the environment completely a priori, but we have an object that will change it's position inside the environment and we expect to know it's location with certain accuracy each time. Second, contrary to the large size room in which those systems are intended to navigate, we expect to estimate the object location in short ranges in which multi-path echoes could be received at the same time creating a superposition difficult to resolve and cause malfunction in those systems [2]. It is our understanding that there are no works related to estimating the position of an object through acoustic analysis under the circumstances described. To achieve the acoustic spatial estimation in this scenario we propose two key steps to the processing pipeline, one in the beamforming stage to increase the spatial discrimination from where the echoes are originated, and second a change in the type of neural network use to infer the location, to employ a network suitable to directly process time series instead of Convolutional Neural Networks. Our purpose is to demonstrate that is possible to locate an object in a small room (e.g. 1m side dimension) using acoustic signal processing with neural networks.

1.2. Document Structure

This document present in **Chapter 1** a brief introduction to the topic and is stated the problem we intent to address with the development of this thesis. **Chapter 2** contains the theoretical background and state of the art in the field. **Chapter 3** details the hypothesis, objectives and scope of the work. **Chapter 4** shows the research proposal and methodology. **Chapter 5** present the schedule for the execution of the proposed work.

Theoretical Framework

In this chapter is described the theoretical concepts needed to understand the work that will be developed during the research's execution.

2.1. Room Impulse Response

An acoustic impulse is sharp and narrow pulse which spectrum is composed by a uniform distribution of all frequencies. The mathematics Dirac delta function $\delta(t)$ is used to represent this kind of impulse. In enclosed spaces the sound waves generated by an impulse will travel and reflect and refract in the wall delimiting the space, for a microphone measuring the environment, it will receive the direct path sound, the sound that travel from the source directly to the microphone, followed by the early reflections and finally a series of pulses called late field reverberation as show in figure 2.1 . The direct path, early reflection and reverberations correspond to the response of the enclosed space to an acoustic impulse and is physically defined by the environmental geometry and materials [29].

The Room Impulse Response (RIR) by definition is the transfer function of a sound wave that travel in the space from it's source to the microphone. The microphones hear the convolution of the emitted sound with the corresponding RIR [8] mathematically showed in equation 1, therefore any sound generated in a point and received in other inside a room is the composition of the original signal plus the RIR of the enviroment.

$$g(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(t)x(t-s)ds \quad (1)$$

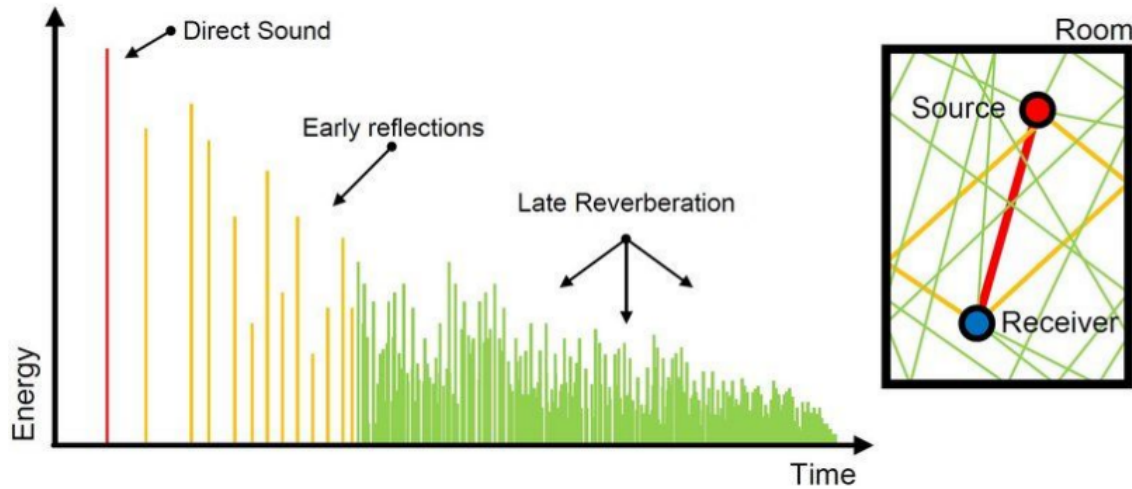


Figure 2.1: Room Impulse Response and the geometrical room representation [26]

2.2. Acoustic Beamforming

In many spatial omnidirectional data transfer context, not only the information carried in the medium is important, the source from where the signal is generated also provide valuable information to properly analyze the incoming data. Beamforming is a spatial filtering technique aims to estimate the direction of the arrival signal inclusive in the presence of noise and other interference in the propagation medium [5]. In acoustic

an array of microphones is used to capture the spatially propagated signals arriving from a certain direction and processes them to investigate the acoustic sources [22].

2.2.1. Microphone Array

A Microphone array combine the signal of multiple omnidirectional microphones in such a way that capture sound from a specific direction while reject sound from other directions [36]. The shape and dimensionality of arrangement depend on the application. The simplest form consist in two microphones spaced by a distance D as show in figure 2.2. The value for D is the minimum distance for a given frequency and its calculation is shown in the equation 2, in the case that the input signals are simple added together for a given frequency f , this arrangement capture sound coming from ± 90 degrees and rejects the sound at 0 and 180 degrees. In a microphone array the higher the number of microphones results in better gain in the desired direction as well as the attenuation of sound coming from the sides of the array [28].

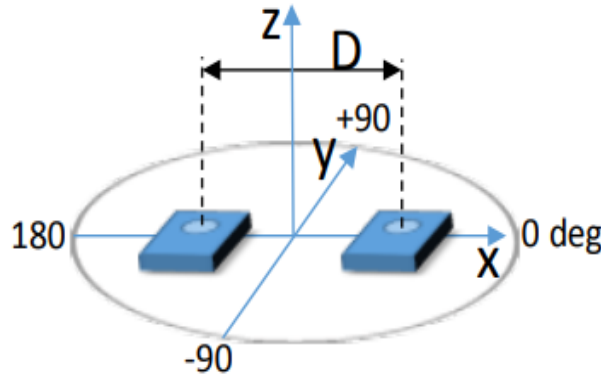


Figure 2.2: Microphone array in binaural arrangement [36]

$$D = \frac{1}{4} \frac{c}{f} \quad (2)$$

2.2.2. Beamforming Pipe

2.2.2.1. Delay-and-Sum Beamforming Technique

Beamforming is based on the physical principle that an acoustic wave originated on a particular location in the space, it travels and arrive at each sensor at different times defined by the distance existing between the unique source and each sensor. The result of adding all these signals depends of the phase relationship between them, so is possible to control the delay for each signal to constructive add the ones from a particular location and, at the same time, destructively cancel the ones originated in other regions [5]. Mathematically the beamforming formula is show in equation 3, where $L(t, S_0)$ is the beamforming output, pm is the microphone output for a signal originated at $S_0 = (x_0, y_0, z_0)$ and captured by a microphone located at S . All the signals are sum together and divided by the total number of microphones M in the array.

$$L(t, s_0) = \frac{4\pi}{M} \sum_{m=1}^M pm(s_0, t + t_0) |S - S_0| \quad (3)$$

2.2.2.2. Deconvolution

In real applications the beamforming output comprehends the acoustic signal convolved with the microphone array response. The deconvolution principle assumes that the recorded signal $r(t)$ is a convolution of the sound $g(t)$ (The result of equation 1) and a transfer function $ma(t)$ which represents the array response to a given source [5]. The Point-Spread-Function (PSF) is the array response to a monopole placed in front of the array. The PSF provides the information required to obtain the array transfer function required to do deconvolution and extract the sound without the distortion created by the array.

2.2.2.3. Post-Filtering

Taking into consideration that the signal measured for each microphone is the sum of the sound and the level of noise, and also that the noises are uncorrelated between each microphone and between the beamforming output signal. The beamforming output can be further enhanced by selectively filter the resulting signal based on the cross-spectral densities between channels in the microphone array [22].

2.3. Time Series Transformation into Images

Given a time series, for the sake of analysis sometimes is desirable to perform certain kind of transformations to the time series that provide a more compact representation of the data or highlight specif properties and attributes of it. Transforming time series into images involves the process of transforming vectors into matrices using a criterion of transformation that map values from one domain to another.

2.3.1. Markov transition Field

In the Markov trasnsition fields (MTF), given a time series $X = (x_1, \dots, x_n)$ of real observations, the serie is discretized based on quantile bins assignation. First the codomine of the function is sub-grouped taking the range of values in X and divided it into Q segments, then each x_i is allocated to bin q_j with $j \in \{1, \dots, Q\}$ if its value is within the segment. Now considering the resulting bin values as observations of a first-order Markov chain and computing the number of occurrences of pairs of back-to-back bins for every pair of bins is possible to represent the ocurrences as a Markov transition matrix of dimension $Q \times Q$. The Markov transition matrix is normalized to transform frequencies into probabilities. then each entry in the matrix is the probability to go from q_j to q_k , for every pair (q_j, q_k) [10].

Finally the dimensionality of the time series is restored by projecting the Markov

matrix $Q \times Q$ into a $n \times n$ matrix called Markov transition field, where each entry for the pair (x_i, x_j) is the probability to go from the bin $q(x_i)$ to bin $q(x_j)$. For a serie of n observations the MTF can formally be defined by the equation 4.

$$\forall i, j \in 1, \dots, n, MTF_{i,j} = \mathbb{P}(q(x_i)|q(x_j)) \quad (4)$$

The figure 2.3 Shows the data trasformations requiered to obtain a MTF, First in (a) the time series is discretized by dividing the variable range in intervals, (b) the Markov transition matrix is calculated and from this the MTF for the probability of each pair in the serie is created as presented in (c).

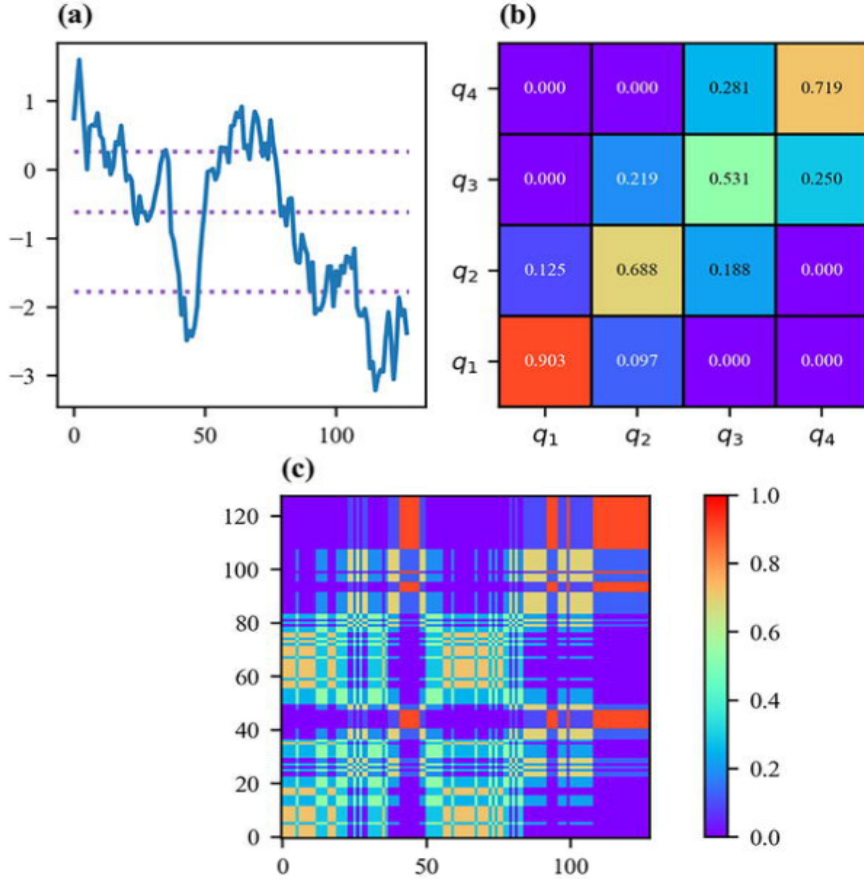


Figure 2.3: Summarized process to build a Markov Transition Field [10]

2.3.2. Gramian Angular Field

The Gramian Angular Field (GAF) is a matrix build on the principle of transform the time series values to the polar domain [35]. Given a time series $X = (x_1, \dots, x_n)$ of real observations, the gramian transformation takes the values x_i and rescale them in the interval $[-1, 1]$ or $[0, 1]$. Then the rescaled values are converted to a polar form considering the rescaled value as the angular cosine and the timestamp the radius. The transformation function is show in 4. From the resulting values only the angle is used as the one that depends of the values of the time series. Two matrices can be build considering the sum or subtraction of pair of angles resulting from 4, the sum is called Gramian Angular Sum Field (GASF), the subtraction of angles receive the name of Gramian Angular Difference Field [11].

$$\begin{aligned} \phi_i &= \arccos(\bar{x}_i), \quad \bar{x}_i \in \bar{X} \\ r_i &= \frac{t_i}{n}, \quad t_i \in \mathbb{N} \end{aligned} \tag{4}$$

2.4. Convolutional Neural Network

A Convolutional Neural Network (CNN) is an artificial neural network architecture used to analyze and learn from visual data presented commonly in the form of images. In its simplest form the structure is composed by three building blocks: convolution layer, pooling layer, and fully connected layer. The figure 2.6 shows and example of how this blocks are connected to analyze an image stage by stages in order to determine the structural condition of an aircraft fuselage.

2.4.1. Convolution operation

Convolution is a common operation used on image filtering to denoise or detect features. A filter also called a convolution kernel is applied to all locations of the image to produce a new filtered image [7]. For a given 2D image X of dimension $I \times J$ with

pixels $x_{i,j}$ a filter W of dimension $K \times L$ with weights $w_{k,l}$ can be convolved to obtain a filtered image $X * W$. The convolution process over the image can be mathematically written as:

$$(w * x)_{i,j} = \sum_{k=-K/2}^{K/2} \sum_{l=-L/2}^{L/2} w_{k,l} x_{i+k,j+l} \quad (5)$$

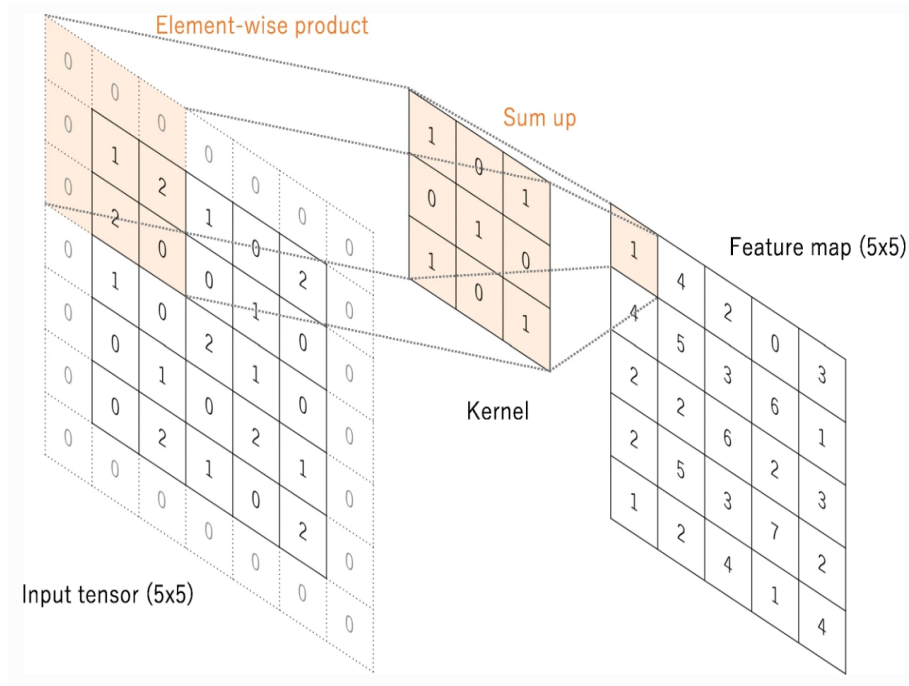


Figure 2.4: Convolution operation over a tensor of 5x5 with zero padding and a stride of 1, resulting in a feature map of 5x5 [40]

The convolution operation in equation 5 does not allow the center of the kernel to overlap the edges of the image, resulting in a reduction in the size of the output feature map. Padding increases the input size by adding extra rows and columns around the outermost elements on each side of the image in order to compensate the dimensional reduction [40]. In practice as shown in figure 2.4 the kernel is applied across the input one step at a time, an element-wise operation is performed in this

location and summed to obtain the output, after this the kernel is moved to the next location to filter other segment of the input. How big is displacement of the kernel is given by the stride, which define the distance between two successive kernel positions.

2.4.2. Pooling

Pooling is a dimension reduction technique where an input map is spatially reduced while keeping the most important features filtered or learned. The pooling operation reduce or collapse a sub region to a smaller dimension helping the learning representation to be more invariant to small length-scale perturbations [15].

Two common pooling techniques are max-pooling and average-pooling, the first one, for the defined pooling window it will consider only the maximum value, in the case of the average pooling is calculated the average of the values inside the window as show in the example of figure 2.5.

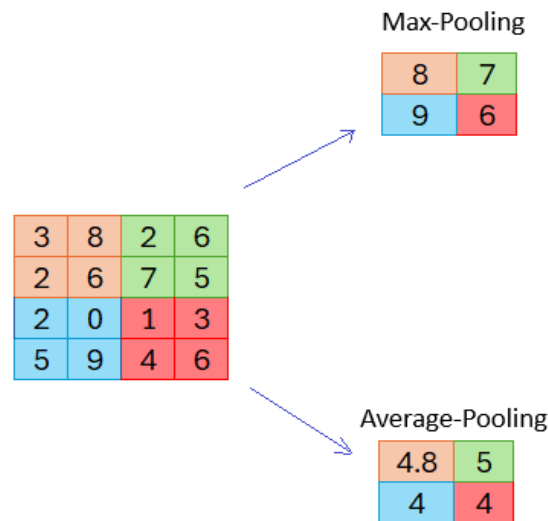


Figure 2.5: Pooling escalation with average pooling and max-pooling

2.4.3. Fully connected layer

The final stage in a CNN is a fully connected layer. This layer is responsible to classify the features extracted from the previous layers. The figure 2.6 shows an example of CNN for aircraft structural health classification with its fully connected layer allocated at the output of the net.

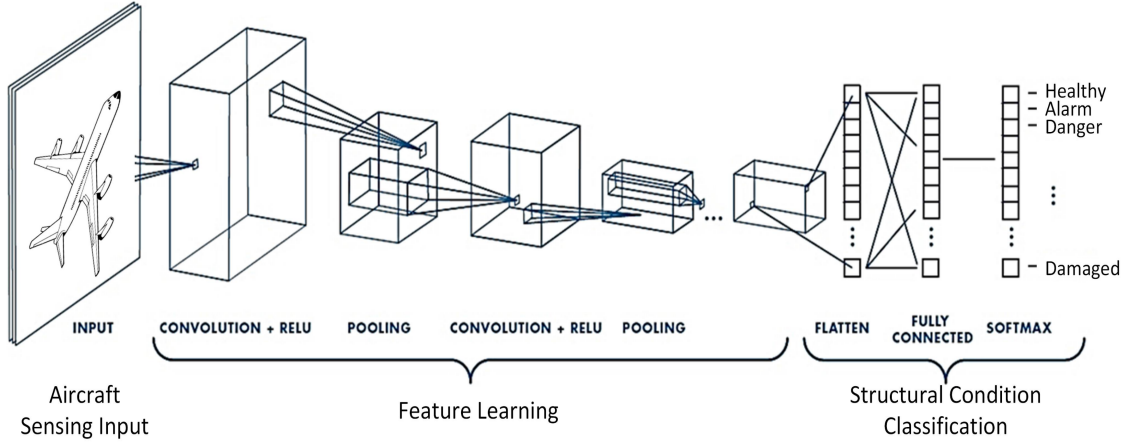


Figure 2.6: Structure of Convolutional Neural Network for aircraft structural health monitoring [31]

2.5. Echo State Network

An Echo State Network (ESN) is a new paradigm in recurrent neural networks called Reservoir Computing (RC). In RC, the reservoir is the recurrent part of the network and has the particularity that it is not trained. Due to the fact that the reservoir skips the training, the network avoids the computationally expensive gradient-descent on recurrent networks, and instead of training the recurrent network, it is defined in such a way that is capable of transforming input signals into a high-dimensional state space values. The state of the reservoir can be processed by a simple fully connected readout network trained to map the reservoir state to a desired output [19].

In the ESN, the reservoir is a randomly created RNN where the weights remain

unchanged. This RNN is exited by an input signal, and its future state is a function of the actual input and the history of the reservoir. The result is generated by training a simple feed-forward network to output the target value. Figure 2.7 [19] shows a comparison of traditional RNN and RC. In RC the bold arrows used to exemplify the weight change resulting from the training process and is only performed at the output layer of the network.

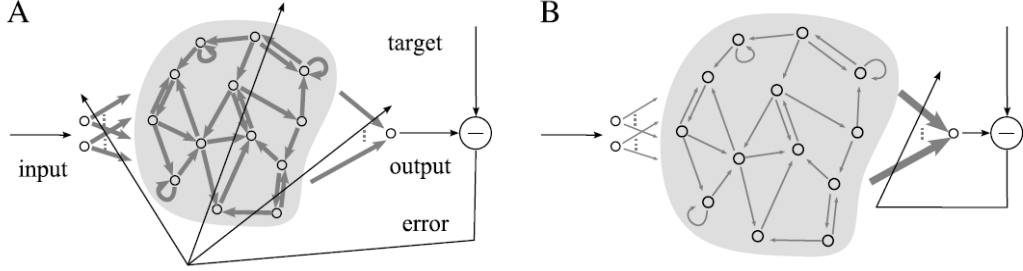


Figure 2.7: Comparison of B) RC vs A) traditional RNN, the bold arrows are weights that change during training [19]

2.5.1. Formalism of ESN

A formal description of an ESN from [30] define an ESN receiving a time series input $U(t) \in R^{D \times 1}$, the state of the reservoir is denoted by $X(t) \in R^{N \times 1}$ and $y(t) \in R^{M \times 1}$ the output. The weights for the stages are: $W_{in} \in R^{N \times D}$ for the input, $W_{res} \in R^{N \times N}$ for the reservoir and $W_{out} \in R^{N \times M}$ for the output. The equations for the internal state of the reservoir and the output are show in 6. The reservoir state depends of the input value at time t and the previous state of the system. The output can be obtained by training to solve a linear regression problem.

$$\begin{aligned} x(t) &= f(W_{in}U(t) + W_{res}x(t-1)) \\ y(t) &= W_{out}x(t) \end{aligned} \tag{6}$$

There exists an important property that the reservoir should meet to ensure the reservoir dynamics is capable of forget the initial transient state [39]. The Echo State Property (ESP) enunciate that a network with state transition equation F has the echo state property if for each input sequence $U = [u(1), \dots, u(N)]$ and all couples of initial states x, x' for the reservoir, it hold the equation 7.

$$||F(U, x) - F(U, x')|| \rightarrow 0 \text{ as } N \rightarrow \infty \quad (7)$$

To ensure the existence of ESP the W_{res} must meet a necessary condition and a sufficient condition as defined as follow:

$$\begin{aligned} ESP &\Rightarrow \rho(W_{res}) < 1 \\ \bar{\sigma}(W_{res}) < 1 &\Rightarrow ESP \end{aligned} \quad (8)$$

Where $\rho(W_{res})$ is the spectral radius parameter of W_{res} and the $\bar{\sigma}(W_{res})$ is the largest singular value of W_{res} . So from equations 8, if $\rho(W_{res})$ and $\bar{\sigma}(W_{res})$ are less than one, the ESP is satisfied and the reservoir dynamic will be correct.

2.6. Related Work

The related work in acoustic processing for object recognition and classification can be divided in two categories based on from where the sound is originated. We can considered a passive sensing approach if the sound stimulus is originated directly in the system, otherwise if the sound is an external stimulus that interact with the

environment before sensed we are under a active sensing approach.

2.6.1. Passive Acoustic Sensing and Analysis

2.6.1.1. Acoustic Imaging

An acoustic image is a sound intensity map that highlight the location of different sound sources in a bi-dimensional plane. An acoustic camera takes this intensity map and superimposed it on a optical image to show the physical source of the sound. As show in figure 2.8 a microphone array is used to ear the sound and a beamforming algorithm process the signals to locate the direction from where the sound is originated.

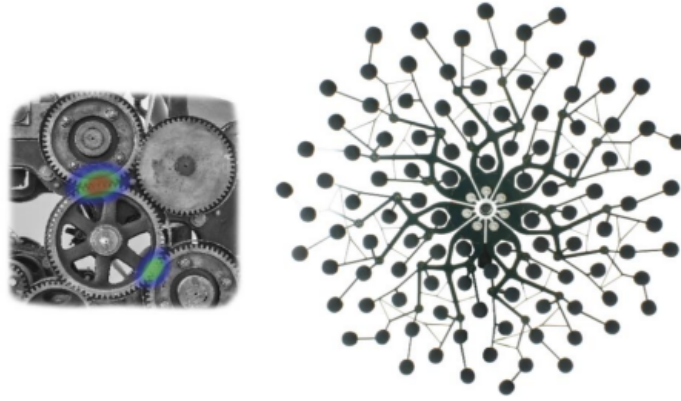


Figure 2.8: Image on the left is an acoustic image, on the right and microphone camera used to capture the image [27]

Acoustic beamforming is a field that has been extensively studied, different shapes and sizes of microphone arrays has been analyzed [28], also exist many techniques to beamforming the signals from the array [22], recently studies are focused on spherical arrays to identify multiple sound sources in a three dimensional space [6; 14; 17].

2.6.1.2. Acoustic Scene Classification

Artificial Intelligence and specially the deep learning subfield powered the development of new techniques to segment, analyze and classify acoustic originated

signals. Acoustic Scene Classification (ASC) use deep learning to learn and distinguish from audio the context in which the recording was performed. The CNN are the first choice for ASC, the required images that are going to be the input to the network are obtained by processing an audio dataset to convert it to spectrograms. Taking the DCASE 2016 dataset, the static Log-Mel spectrograms and dynamic first derivative spectrograms are used in [4] to build the two channels of the images that will be used as input training of the CNN. Han et al. in [13] used the DCASE 2017 dataset to improve the input data to train, this through multiple versions of mel-spectrograms divided in binaural representation, source separation and background subtraction. In binaural representation instead of mono audio representation they used left-right and mid-side spectrograms. The source separations divide the dataset into two using harmonic-persecutive sound separation algorithms to increase the dataset to get the representation in armonic and percusive sound. Finally media filtering is used in the spectrograms to steady noise from the environment. The resulting multiple version spectrograms are used to training a ConvNet architecture of 8 convolution layers inspired in a VGG net. They obtained a mean accuracy improvement compared with the baseline used in the work and a final 74.8% to a 80.5% accuracy with the various prepossessing methods previously described. The work presented in [34] used the DCASE 2019 dataset and the log-mel spectrograms to training two deep learning algorithms to perform the classification, a CNN but also a Convolutional Recurrent Neural Network (CRNN) are the architectures choosed in this work. The highest recognition obtained was for the CRNN with 90.96% of accuracy, a value superior to 73.61% of the CNN used as baseline. A different objective is presented by Byttebier in [3] where small foodprint neural network models are investigated. The author compared several small neural networks employing the DCASE 2020 to evaluate them and found that a 16-bit CNN with squeeze excitation blocks, regularized with MIX UP and weight decay is the one that shows the best

accuracy of 66.84% with only 90 kilobyte of parameters.

2.6.1.3. Acoustic Object Recognition

Acoustic object recognition aims to recognize an object in base of the sound that emits whether it is self-generated as the sound produced by and air-conditioner during normal operation or external, in response to a stimulus like the characteristic sound produced by the wood when someone is knocking a door. For the work presented by Luo et al. in [20] a dataset was built using 30 different objects, each of them was hit by a marker pen to generate a characteristic sound that was recorded. A training set of 100 hits (in different locations of the object) per object was created, and an extra 20 hits per object were obtained as well in order to be use in the test phase. With the dataset a denoise-autoencoder was trained to recognize the different sound of the object from which its originated. The recognition rate for the autoendoders was 91.50%, a value superior to the baseline provided by a Support Vector Machine with recognition rate of 82%.

Other dataset was created by Tran et al. in [33], in this case the input were LEGO objects that fall from 30cm high. The sound of the LEGO pieces hitting the ground was used to create spectrograms to training a CNN called LegoNet in order to recognize the LEGO pieces by the sound of its free-fall impact, LegoNet achieved an accuracy of 99.21%, a value superior to the other networks used for comparison. For robotic perception in [16] was develop a framework in which a robot can recognize the object inside a container by shaking the container and analyzing the sound. An kernel K-nearest neighbor algorithm in a open environment is used to shows the feasibility of the method. A generative physical, audio-visual synthesis pipeline is presented in [41], the model takes as input the shape of an object, its material and one scene configuration, a physical engine simulate motion and collision for that particular object. An audio engine takes the shape, material and the collision information to synthesize the sound.

The database generated is used in the training a two-stream CNN, one ResNet-18 for the image data, together with a SoundNet-8 for the audio data. The net demonstrated that the knowledge learned in the synthetic audio-visual dataset was transferred to real world data once it was used to infer object materials from auditory data, and estimate shape attributes from audio-visual data.

2.6.2. Active Acoustic Sensing and Analysis

2.6.2.1. RIR for Room Inference

For an enclosed space, the RIR provided a unique representation of the geometry and material inside it, sound like music and speech recorded in rooms comprehend this echoes information and can be use to recognize the room in which the recording was performed. This idea is presented in [24] where the authors developed a machine learning system to identify rooms for the reverberant audio samples recorded in those rooms and achieved and 85% accuracy in the task. The work in [1] use the RIR and speech data to train a model to infer the room volume from a single microphone recorded sound, with this information the model with an accuracy of 85.8% classified the rooms into three categories: small, large or hall. Using microphone arrays and analyzing the RIR obtained from them in [9] the authors are able to estimate the room geometry by exploring the properties of Euclidean distance matrices and show that under certain conditions the first order echoes provide a unique description of the geometry of convex polyhedral rooms. In [18] an indoor positioning device is presented using the RIR as the "fingerprint" to identify the room in which the system is located, the accuracy varies depending of the type of room, ranging from 65% in the identification of living rooms to 95% respect to bathrooms. The work in [37] shows a dataset of RIR in which people was standing in the rooms and relocated to different positions inside it to generate samples. The author shows the potential of using machine learning approaches over the dataset for tasks related to localizing, identifying and detect humans in those rooms.

2.6.2.2. Acoustic Object Shape Estimation

Acoustic perception of shapes are explored in [21], this work proposes the use of acoustic standing waves to recognize the size and shape of a target object analyzing the distance spectrum profile of the standing waves created by the interference between transmitted and reflected sounds, In this work Manabe et al. were capable of estimate the dimensions of a rectangular board of 1.2m x 0.9m and the mean absolute error of the estimation was 0.0194m, with this result they demonstrated that the spectrum profile encodes information of the target and can be use to estimate the shape of an object. The approach presented by Brunetto et al. in [2] result in the developed of the Batvision system capable of see deep spacial layouts by listening to echos recorded by two microphones as artificial ears. For Batvision a dataset is created using mobile robots that generate acoustic chirps signals at the same time a deep image is captured, with this dataset of pairs audio-image an adversarial discriminator is trained to generate deep images based on sound chirps. The system was capable of predict detailed indoor scenes and obstacles with a L1 loss as low as 0.072, sometimes even outperforming a deep mapping created by a stereo vision algorithm. Following the findings in Batvision the work presented in CatChatter paper [32] shows two separate results, first an improved version of Batvision is developed in which an input encoder from the model was removed, this encoder was processing the acoustic spectrogram before reaching the adversarial discriminator and was causing that spatial inputs from the spectrogram were lost at the time it reach to the U-Net decreasing the inference capacity of the model. The other result presented in [32] was from a new system developed that use lidar to create 3D, 360 degrees depth images together with an increase in the acoustic perception from 2 microphones to 4 in order to capture echoes from all the directions. The results obtained by this new method are similar to the ones reported in Batvision [2] but the field of prediction due to 360 deep information and the extra 2 microphones was 4 times bigger than the obtained with deep camera solution and two microphones system. To detect

basic shapes and deformation acoustic sensing was used in [25] by Rothstein et al. to identify and monitor small hollow object from inside. The deformation of the shapes are created by generating similar shapes with alter dimensions or by blocking a certain part of the object interior. A square wave sound signal is emitted in the interior of the objects and a 4 microphone array records the echos produced in the interior. Acoustic features are created from these signals and a classifier is trained to distinguish the shape and deformation of the object under study. Their work presented that for the objects the shape can be identify with 80% accuracy, the existence of deformation with 100% accuracy and localize the deformation with 90% accuracy.

Hypothesis and Objectives

3.1. Hypothesis

The compendium of work presented in the Chapter 2 demonstrated that is possible detect with precision the source of a sound in a tridimensional space, recognize a particular room by its unique acoustic and even create a deep image of segments of this room using only audible sound echoes and infer shapes of object based on acoustic features. Those results support the idea that for a given object inside a enclosed space is possible to learn a function that map the RIR to the location of the object in that space. Based on these ideas our hypothesis is:

"An algorithm pipeline with beamforming and ESN that utilize the rich spatial-geometrical information comprehended in a set of RIR can infer the location and orientation of a cube of 20cm side allocated inside an small enclosed space of near of $1m^3$ ¹, this with a movement distance resolution of at least 5cm and achieve less

¹Region with acoustic superposition found by [2]

than 2cm of error in position and 10 degrees in orientation² with an error less than 6 degrees in the estimation from two consecutive positions."

3.2. Objectives

The current research established the following objectives:

3.2.1. Main Objective

Develop and evaluate a processing pipeline using beamforming together with CNN and ESN to transform a multichannel RIR to spatial coordinates and angles that situate an object within an enclosed space.

3.2.2. Specific Objectives

1. Collect a dataset of acoustic responses of an object positioned in different locations inside a room.
2. Develop a method using CNN to learn from the dataset a function that estimate the coordinates and orientation of an object inside the space.
3. Develop a method using ESN to learn from the dataset a function that estimate the coordinates and orientation of an object inside the space.
4. Build a beamforming stage that divide the measure space in eight regions of acoustic sources.
5. Evaluate experimentally the accuracy of the estimators.

²Based on the size of the objects and maximin deformation presented in [25]

3.2.3. Deliverables

Specific Objective 1: A dataset will be delivered containing acoustic recordings of echoes in the form of time series and its corresponding object location-orientation.

Specific Objective 2: Thesis document with detailed description of the concepts, observation and results analysis that justify the system design with CNN.

Specific Objective 3: Thesis document with detailed description of the concepts, observation and results analysis that justify the system design with ESN.

Specific Objective 4: Thesis document with detailed description of the concepts, observation and results analysis that justify the beamforming.

Specific Objective 5: Thesis document with analysis and evaluation of the proposed solutions considering performance metrics under experimental operation.

3.3. Scope and Limitations

Scope and limitation 1: The development of the measure box and the collection of the dataset will be part of the work.

Scope and limitation 2: Only two ML approaches will be considered: CNN as the ground truth and ESN as an optimal solution.

Scope and limitation 3: The proposed system will not work on real time, even for the experimental evaluation the data will be first collected and later presented to the model in order to obtain the evaluation metrics.

Scope and limitation 3: The programming language will be python, the hardware to process the data will be a desktop personal computer.

Research Procedure

4.1. Research Design

The following chapter details the physical setup to develop the measurements that constitute the dataset under study, also present a description of testbench and processing pipeline in which the data will flow from the acoustic inputs towards the desire location-orientation output.

4.2. Methodology

The methodology starts with the data collection under an experimental setup, the data will be curated to eliminate random noise and presented as a time series dataset. The resulting dataset will be used to train and test four neural network pipelines. The predicted location-orientation error is calculated and a multifactor Analysis of Variance (ANOVA) is proposed as the main criteria to evaluate the significant variance of the error of each processing pipeline used to estimate the location and orientation of an

object in a room.

4.2.1. Experiment Design

The research require the development of a manipulable and fully controllable environment in which the acoustic measurements are going to be performed. A small box type room is the volumetric cubic space proposed as the experimental container. The interior of the room needs to be easily configurable, for this a robotic arm will be allocated inside to manipulate object in the interior. An acoustic signal originated at one corner of the room will be measured and captured by microphones located in the opposite sides of the room. The figure 4.1 shows the measurement setup, a six DOE robotic arm located in the center is used to move and change the position and orientation of one cube inside the box. The blue circles correspond to three microphones arrays placed in the sides of the room. One speaker positioned in the opposite angle to side in which the microphones are located is the source of the acoustic signal that will be recorded. In yellow is show the trajectory for the direct sound. The arrangement is intended to have a three axis capture composition.

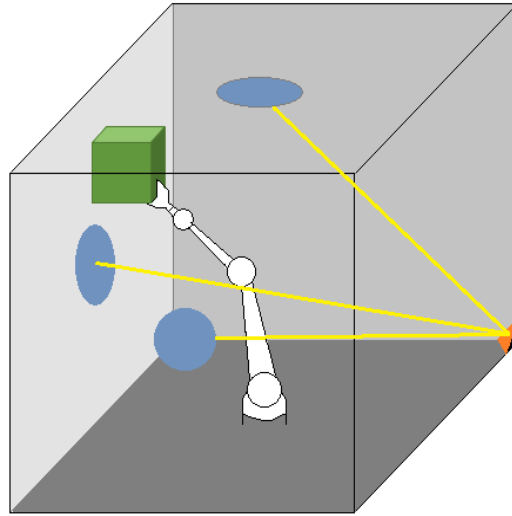


Figure 4.1: Measure box with: the robotic arm, the microphone measurement arrays and the source signal speaker.

Each microphone array is composed of four microphones placed in pairs as show in figure 4.2. A binaural horizontal pair allowed directive selection in X axis, while a binaural vertical does it in Y axis. The figure 4.3 shows the different pipelines proposed to transform the echoes measurements into coordinates.

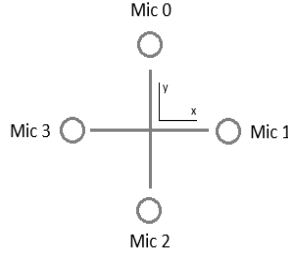


Figure 4.2: Microphone array configured as binaural pairs in horizontal and vertical

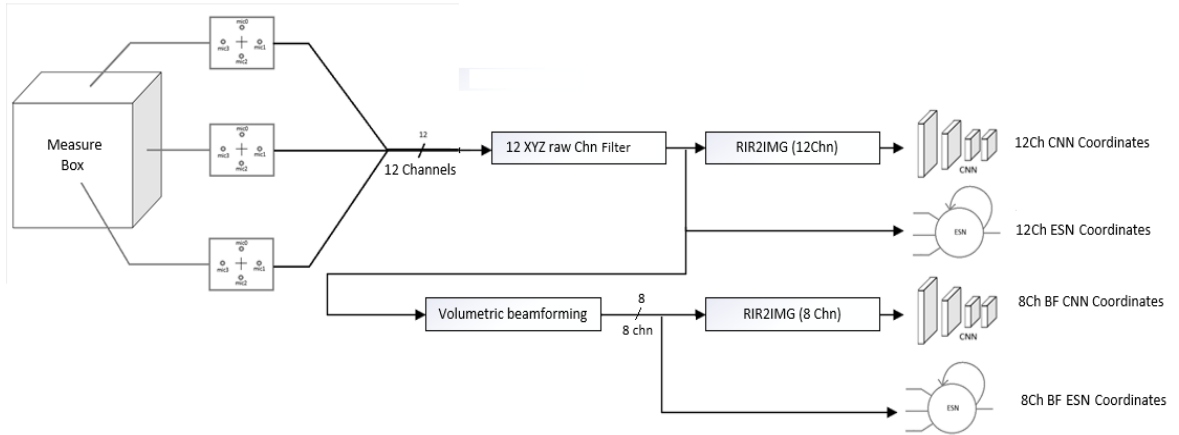


Figure 4.3: Acoustic to location processing pipeline

In figure 4.3 the three arrays together result in a twelve channel measure, this channels are used to feed the pipelines. First a filtering stage is intended to remove the noise based on the cross-spectral densities between channels [22]. The post filter twelve channels are feeding four parallel processing paths. One that use only CNN needs the time series to be transformed to images representation in order to train and evaluate the performance of a CNN as the coordinates estimator. Exist other that use ESN

and due the fact that the recurrent neural network by principle can directly learn from time series, we explore the impact that have the time series to image transformation by removing this stage and train directly a RC RNN. Finally An improvement suggested to overcome the superposition problem consider the use of beamforming to create a segmented origin representation of the acoustic reverberation, with this the room space is divide in 8 different regions and the sound originated from those regions is contained in a group of new created channel that will also feed a CNN and ESN estimators.

4.2.2. Factors and Levels

For the experimental analysis, the factors and their respective levels to consider are summarized in the table 4.1. As described in previous section the microphone location will have three different arrays located in different wall of the room. The channel count is twelve. The beamforming effect is also considered. The Network is a factor as well with two proposed topologies CNN and ESN as levels.

Table 4.1: Factors and levels for the experiment.

	Factors		
	Mic location - Num Channels	Beamforming	Network
Levels	Three locations, 12 Channels	with without	CNN ESN

The minimal amount of repetitions required by an ANOVA analysis is two, for this experiment we are going to perform 5 runs with 500 points measurements each run.

4.2.3. Response Variable

The response variable is the error from the estimated coordinates of the XYZ location and two angular values that describe the orientation of the cube in the room. Those values are obtained from the output of the neural networks.

4.2.4. Data Collection

The data will be collected on six rounds of measurement as show in table 4.2, a first sample of 1000 values responses correspond to train the neural networks, the other five repetitions of 500 samples are going to be use for the experiment analysis. For the data recollection a python script will be responsible to control the arm to move the object to a new location, trigger the acoustic swap signal and open the recording microphones channels to record the response. The start location will be randomly selected, a measure will be perform followed by a small local random increments less than 5 cm to calculate the new location. after 10 small measurements, a new full random location will be calculated and the robotic arm will be relocated to that position and a small random displacement again will be perform. This process will continue until the sample for the run is completed.

Table 4.2: Acoustic Samples

Acoustic Measurements						
	Type Per Use					
	Training	test1	test2	test3	test4	test5
Samples	1000	500	500	500	500	500

4.2.5. Statistical Tool Used Description

All the data analysis will be perform in R using the package extension for the multifactor ANOVA processing.

Schedule

The figure 5.1 shows the planned schedule, the duration of the investigation comprehends two semesters. The development of the writing document will be covered progressively during all the project.

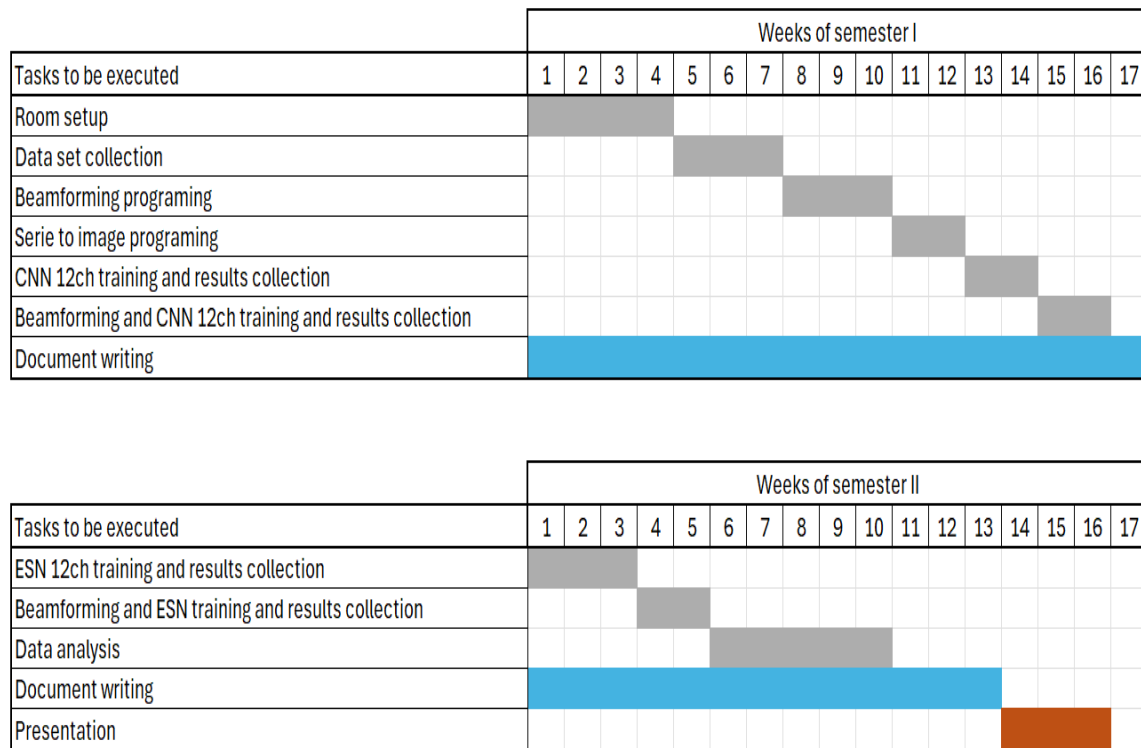


Figure 5.1: Schedule

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